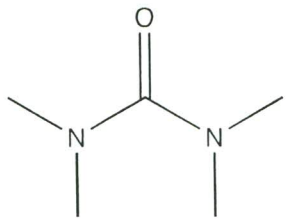
Cume Exam: Protein Structure

1. When a protein is thermally denatured in aqueous solution, it becomes less water-soluble and often precipitates from solution. Speculate as to why this happens?

2. The polypeptide, poly-L-ARG, exists in a random coil conformation @ pH 8. As the pH increases above 11, the polypeptide undergoes a transition to an α-helix conformation. How can this transition be explained?

3. The molecule urea causes denaturation of 2° and 3° structure of proteins, but does not affect the 1° structure of the protein. Study the structure of urea, given below, and explain how urea could disrupt the 2° and 3° structure of protein, but not destroy the 1° structure of the protein.